

# Reconstruction of the initial functions from the solutions of the fractional wave equation on the light cone trace

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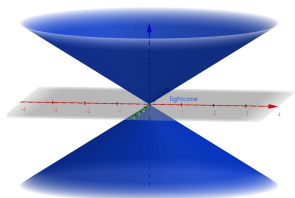
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# Introduction

## light cone

- (a) The path in all directions through which the light can travel along space-time.
- (b) Satisfies the wave equation because light is a wave.
- (c) Can express light cone as  $t = \pm|\mathbf{x}|$  and here we deal with  $t = |\mathbf{x}|$  case.



time axis : one second  
spatial axes : 300,000 km

Figure: light cone.

- The fractional wave equation

$$\begin{aligned} D_t^\alpha p(\mathbf{x}, t; \alpha) &= -(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} p(\mathbf{x}, t; \alpha), \\ (\mathbf{x}, t) &= (x_1, x_2, t) \in \mathbb{R}^2 \times (0, \infty), \quad 1 < \alpha \leq 2 \end{aligned} \quad (1)$$

- The Riesz space-fractional derivative of order  $\alpha$

$$\mathcal{F}(-(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} f)(\boldsymbol{\xi}) := -|\boldsymbol{\xi}|^\alpha (\mathcal{F}f)(\boldsymbol{\xi})$$

where  $\mathcal{F}$  : the Fourier transform

- The Caputo time-fractional derivative of order  $\alpha$

$$(D^\alpha h)(t) := (I^{m-\alpha} h^{(m)})(t), \quad m-1 < \alpha \leq m, \quad m \in \mathbb{N},$$

where  $I^\alpha$ : the Riemann-Liouville fractional integral

$$(I^\alpha h)(t) := \begin{cases} \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} h(\tau) d\tau, & \text{if } \alpha > 0, \\ h(t), & \text{if } \alpha = 0 \end{cases}$$

Suppose  $f \in C_c^\infty(\mathbb{R}^2)$ .

Solution of 
$$\begin{aligned} D_t^\alpha p(\mathbf{x}, t; \alpha) &= -(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} p(\mathbf{x}, t; \alpha) \\ p(\mathbf{x}, 0; \alpha) &= f(\mathbf{x}), \text{ and } D_t p(\mathbf{x}, t; \alpha)|_{t=0} = 0 \end{aligned} :$$

$$p(\mathbf{x}, t; \alpha) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i\xi \cdot \mathbf{x}} E_\alpha(-|\xi|^\alpha t^\alpha) \mathcal{F}f(\xi) d\xi, \quad (2)$$

where  $E_\alpha$  : the Mittag-Leffler function

$$E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(1 + \alpha k)}, \quad \operatorname{Re}(\alpha) > 0, \quad z \in \mathbb{C}. \quad (3)$$

$$p(\mathbf{x}, t; \alpha) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i\xi \cdot \mathbf{x}} E_\alpha(-|\xi|^\alpha t^\alpha) \mathcal{F}f(\xi) d\xi.$$

- Define the linear operator  $\mathcal{W}_\alpha$

$$\mathcal{W}_\alpha f(\mathbf{x}) = p(\mathbf{x}, |\mathbf{x}|; \alpha). \quad (4)$$

**Goal** : reconstruction  $f(\mathbf{x})$  from  $p(\mathbf{x}, |\mathbf{x}|; \alpha)$

When the spatial dimension is odd, Rakesh and Yuan have already provide the inverting the operator trace the solution on the light cone from the initial function [4].

**Strategy**

$$\mathcal{W}_\alpha f \longmapsto (\mathcal{W}_\alpha f)_m \longmapsto (\mathcal{F}f)_m \longmapsto \mathcal{F}f \longmapsto f$$

where  $(\mathcal{W}_\alpha f)_m$  : Fourier coefficient of  $\mathcal{W}_\alpha f$

# Fourier series

Fourier series of  $\mathcal{W}_\alpha f$  and  $\mathcal{F}f$

$$\mathcal{W}_\alpha f(r_{\mathbf{x}}\boldsymbol{\theta}_{\mathbf{x}}) = \sum_{m=-\infty}^{\infty} (\mathcal{W}_\alpha f)_m(r_{\mathbf{x}}) Y_m(\boldsymbol{\theta}_{\mathbf{x}}) \quad \text{and} \quad (5)$$

$$\mathcal{F}f(r_{\boldsymbol{\xi}}\boldsymbol{\theta}_{\boldsymbol{\xi}}) = \sum_{m=-\infty}^{\infty} (\mathcal{F}f)_m(r_{\boldsymbol{\xi}}) Y_m(\boldsymbol{\theta}_{\boldsymbol{\xi}}), \quad (6)$$

Here  $Y_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}$ ,

$(r_{\mathbf{x}}, \boldsymbol{\theta}_{\mathbf{x}}) \in [0, \infty) \times \mathbb{S}^1$  with  $\mathbf{x} = r_{\mathbf{x}}\boldsymbol{\theta}_{\mathbf{x}}$ , where  $\boldsymbol{\theta}_{\mathbf{x}} = (\cos \theta_{\mathbf{x}}, \sin \theta_{\mathbf{x}})$ ,  
 $(r_{\boldsymbol{\xi}}, \boldsymbol{\theta}_{\boldsymbol{\xi}}) \in [0, \infty) \times \mathbb{S}^1$  with  $\boldsymbol{\xi} = r_{\boldsymbol{\xi}}\boldsymbol{\theta}_{\boldsymbol{\xi}}$ , where  $\boldsymbol{\theta}_{\boldsymbol{\xi}} = (\cos \theta_{\boldsymbol{\xi}}, \sin \theta_{\boldsymbol{\xi}})$ .

# Mellin transform

Let  $f(x)$  be a locally integrable function on  $(0, \infty)$

$$\mathcal{M}f(s) = \int_0^{\infty} x^{s-1} f(x) dx, \quad s \in \mathbb{C}. \quad (7)$$

If

$$f(x) = \begin{cases} O(x^{-a-\epsilon}), & \text{as } x \rightarrow 0^+, \\ O(x^{-b+\epsilon}), & \text{as } x \rightarrow +\infty, \end{cases} \quad \epsilon > 0, \quad a < b \quad (8)$$

then the integral (7) converges absolutely and defines an analytic function in the strip  $a < \operatorname{Re}(s) < b$ .

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} t^{-s} \mathcal{M}f(s) ds, \quad a < c < b \quad (9)$$

Property of Mellin transform :

$$\begin{aligned} \mathcal{M}(f \times g) &= \mathcal{M}f \cdot \mathcal{M}g, \\ f \times g(t) &= \int_0^{\infty} f(\tau) g\left(\frac{t}{\tau}\right) \frac{d\tau}{\tau}. \end{aligned}$$

# Main results

## Strategy

$$\mathcal{W}_\alpha f \longmapsto (\mathcal{W}_\alpha f)_m \longmapsto (\mathcal{F}f)_m \longmapsto \mathcal{F}f \longmapsto f$$

- The relation between  $(\mathcal{F}f)_m$  and  $(\mathcal{W}_\alpha f)_m$

## Theorem 1

For  $f \in C_c^\infty(\mathbb{R}^2)$ ,  $(\mathcal{W}_\alpha f)_m(r) = \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha r^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi r) d\rho_\xi$ ,

where  $J_\nu(\cdot)$  : Bessel function (of the first kind) of order  $\nu$ .

- To use Mellin transform

$$r^{-2}(\mathcal{W}_\alpha f)_m(r^{-1}) = \frac{(-1)^{|m|} i^m}{2\pi} (\mathcal{F}f)_m \times E(r; \alpha, m), \quad (10)$$

where  $E(r; \alpha, m) = E_\alpha(-r^{-\alpha}) J_{|m|}(r^{-1}) r^{-2}$ .

## Theorem 2

Let  $f \in C_c^\infty(\mathbb{R}^2)$ . For  $\frac{3}{2} + \epsilon < c < 2 + |m| - \epsilon$  with  $\epsilon < \frac{1}{4}$ , we have

$$(\mathcal{F}f)_m(r) = i^{m-1} \int_{c-i\infty}^{c+i\infty} r^{-s} \frac{\mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

# Theorem 1

For  $f \in C_c^\infty(\mathbb{R}^2)$ .  $(\mathcal{W}_\alpha f)_m(r) = \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha r^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi r) d\rho_\xi$ ,

where  $J_\nu(\cdot)$  : Bessel function (of the first kind) of order  $\nu$ .

## Proof

$$\begin{aligned}\mathcal{W}_\alpha f(\mathbf{x}) &= \frac{1}{(2\pi)^2} \int_{\mathbb{S}^1} \int_0^\infty e^{i\rho_\xi \theta_\xi \cdot \mathbf{x}} E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) \mathcal{F}f(\rho_\xi \theta_\xi) \rho_\xi d\rho_\xi dS(\theta_\xi) \\ &= \frac{1}{(2\pi)^2} \sum_{m=-\infty}^\infty \int_0^\infty E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi \left( \int_{\mathbb{S}^1} e^{i\rho_\xi \theta_\xi \cdot \mathbf{x}} Y_m(\theta_\xi) dS(\theta_\xi) \right) d\rho_\xi \\ &= \sum_{m=-\infty}^\infty \frac{i^m}{2\pi} \int_0^\infty E_\alpha(-\rho_\xi^\alpha |\mathbf{x}|^\alpha) (\mathcal{F}f)_m(\rho_\xi) \rho_\xi J_m(\rho_\xi |\mathbf{x}|) d\rho_\xi Y_m\left(\frac{\mathbf{x}}{|\mathbf{x}|}\right),\end{aligned}$$

The Funk-Hecke theorem in [2, p 198] :

$$\int_{\mathbb{S}^1} e^{i\rho_\xi |\mathbf{x}| \theta_\xi \cdot \frac{\mathbf{x}}{|\mathbf{x}|}} Y_m(\theta_\xi) dS(\theta_\xi) = (2\pi) i^m J_m(\rho_\xi |\mathbf{x}|) Y_m\left(\frac{\mathbf{x}}{|\mathbf{x}|}\right).$$

# Well - defined

- to Use the mellin transform

$$r^{-2}(\mathcal{W}_\alpha f)_m(r^{-1}) = \frac{(-1)^{|m|} i^m}{2\pi} (\mathcal{F}f)_m \times E(r; \alpha, m),$$

where  $E(r; \alpha, m) = E_\alpha(-r^{-\alpha})J_{|m|}(r^{-1})r^{-2}$ , beacuse  $J_m = (-1)^{|m|}J_{|m|}$ .

Since  $f \in C_c^\infty(\mathbb{R}^2)$ ,  $\mathcal{F}f$  decays quickly and so  $(\mathcal{F}f)_m$  still decays quickly.  $\mathcal{M}((\mathcal{F}f)_m)(s)$  and inversion formula for  $\mathcal{M}((\mathcal{F}f)_m)(s)$  are well defined for  $0 + \epsilon < \operatorname{Re}(s)$ .

## Lemma

If  $E(r; \alpha, m) = E_\alpha(-r^{-\alpha})J_{|m|}(r^{-1})r^{-2}$ , then we have

$$E(r; \alpha, m) = \begin{cases} O(r^{-(\frac{3}{2})}), & \text{as } r \rightarrow 0^+, \\ O(r^{-(2+|m|)}), & \text{as } r \rightarrow +\infty, \end{cases}$$

so  $\mathcal{M}(E)(s; \alpha, m)$  and inversion formula for  $\mathcal{M}(E)(s; \alpha, m)$  are well defined for  $\frac{3}{2} + \epsilon < \operatorname{Re}(s) < 2 + |m| - \epsilon$  with  $\epsilon < \frac{1}{4}$ .

## Lemma

If  $E(r; \alpha, m) = E_\alpha(-r^{-\alpha})J_{|m|}(r^{-1})r^{-2}$ , then we have

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so  $\mathcal{M}(E)(s; \alpha, m)$  and inversion formula for  $\mathcal{M}(E)(s; \alpha, m)$  are well defined for  $\frac{3}{2} + \epsilon < \operatorname{Re}(s) < 2 + |m| - \epsilon$  with  $\epsilon < \frac{1}{4}$ .

## Proof

$E(r; \alpha, m) = O(r^{-(\frac{3}{2})})$  as  $r \rightarrow 0^+$ .

- $E_\alpha(-r^{-\alpha}) = O(r^0)$  as  $r \rightarrow 0^+$
- $J_\nu(x) \approx x^{-\frac{1}{2}} \cos\left(x - \frac{(2\nu+1)\pi}{4}\right)$  for large  $x$ , so  $J_{|m|}(r^{-1}) = O(r^{\frac{1}{2}})$  as  $r \rightarrow 0^+$

$E(r; \alpha, m) = O(r^{-(2+|m|)})$  as  $r \rightarrow +\infty$ .

- $E_\alpha(-r^{-\alpha})$  has finite value as  $r \rightarrow +\infty$ , so  $E_\alpha(-r^{-\alpha}) = O(r^0)$  as  $r \rightarrow +\infty$
- $J_\nu(x) \approx \frac{x^\nu}{2^\nu \Gamma(\nu+1)}$  for  $x$  near 0, so  $J_{|m|}(r^{-1}) = O(r^{-|m|})$  as  $r \rightarrow +\infty$

## Theorem 2

Let  $f \in C_c^\infty(\mathbb{R}^2)$ . For  $\frac{3}{2} + \epsilon < c < 2 + |m| - \epsilon$  with  $\epsilon < \frac{1}{4}$ , we have

$$(\mathcal{F}f)_m(r) = i^{m-1} \int_{c-i\infty}^{c+i\infty} r^{-s} \frac{\mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

## Proof

$$\begin{aligned} \mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s) &= (-1)^{|m|} i^m (2\pi)^{-1} \mathcal{M}[(\mathcal{F}f)_m \times E](s; \alpha, m) \\ &= (-1)^{|m|} i^m (2\pi)^{-1} \mathcal{M}((\mathcal{F}f)_m)(s) \cdot \mathcal{M}(E)(s; \alpha, m). \end{aligned}$$

Taking the Mellin inverse formula (9) of

$$\mathcal{M}((\mathcal{F}f)_m)(s) = \frac{i^m (2\pi) \mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)},$$

we have

$$(\mathcal{F}f)_m(t) = i^{m-1} \int_{c-i\infty}^{c+i\infty} t^{-s} \frac{\mathcal{M}[(\cdot)^{-2}(\mathcal{W}_\alpha f)_m(\cdot^{-1})](s)}{\mathcal{M}(E)(s; \alpha, m)} ds.$$

## Case $n = 3$

$Y_{lk}(\theta, \varphi)$  the spherical harmonics, which form an orthonormal basis in  $L^2(\mathbb{S}^2)$

$$Y_{lk}(\theta, \varphi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-|k|)!}{(l+|k|)!}} P_l^{|k|}(\cos \theta) e^{ik\varphi}, \quad k = -l, \dots, l, \quad l = 0, 1, 2, \dots$$

where  $\theta$  : polar angle

$\varphi$  : azimuthal angle

$P_l^k$  : associated Legendre functions

$$\left( (1-t^2)(P_l^k)''(t) - 2t(P_l^k)'(t) + \left\{ l(l+1) - \frac{k^2}{1-t^2} \right\} P_l^k(t) \right) = 0$$

In three dimensions,  $f$  can be reconstructed from  $\mathcal{W}_\alpha f$  in the same way as reconstructed in two dimensions.

**In all dimensions, we can reconstruct  $f$  from  $\mathcal{W}_\alpha f$ .**

# Reference



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# Thank you.